

OBJECT ORIENTED PROGRAMMING

#3

MUNKHZUL TSOGOO

EXERCISE #4

```
#include <iostream>

using namespace std;

class Person {
public:

    string m_name;
    string m_gender;
    int m_age;
    string m_email;

    int introduceMyself() {
        cout << "My name is " << m_name << ".\n";
        cout << "I am " << m_gender << " and " << m_age << " years old.\n";
        cout << "My email address is " << m_email << ".\n";
        return 0;
    }
};

int main() {
    Person ryu = { "Victor Ryu", "male" , 30, "IGS2130v1@inha.ac.kr" };
    Person laura = { "Laura Lee", "female" , 23, "IGS2130v2@inha.ac.kr" };

    ryu.introduceMyself();
    cout << endl;
    laura.introduceMyself();

    return 0;
}
```

- **This code uses class. The class has member variables and a member function inside it, and it is all public. Then the main function makes 2 classes with its data in the**

parameter so it assigns the values to the member variables according to its order that has been written in the parameter. Then the main function calls on the member function “introduceMyself” from the “Person” class.

```
My name is Victor Ryu.  
I am male and 30 years old.  
My email address is IGS2130v1@inha.ac.kr.  
  
My name is Laura Lee.  
I am female and 23 years old.  
My email address is IGS2130v2@inha.ac.kr.
```

EXERCISE #5

```
#include <iostream>  
using namespace std;  
  
class Number {  
    double m_n1;  
  
public:  
    double setValue(double s) {  
        m_n1 = s;  
        return m_n1;  
    }  
  
    double getValue() {  
        double get = m_n1;  
        return get;  
    }  
  
    double add(double a) {  
        m_n1 += a;  
        return m_n1;  
    }  
  
    double sub(double b) {  
        m_n1 -= b;  
        return m_n1;  
    }  
  
    double mul(double c) {  
        m_n1 *= c;  
        return m_n1;  
    }  
  
    double div(double d) {  
        m_n1 /= d;  
        return m_n1;  
    }  
};
```

```

int main() {
    Number n1, n2;
    n1.setValue(0.0);
    cout << "n1 : " << n1.getValue() << endl;
    cout << "n1 += 10.5 : " << n1.add(10.5) << endl;
    cout << "n1 -= 1.5 : " << n1.sub(1.5) << endl;
    cout << "n1 *= 3.0 : " << n1.mul(3.0) << endl;
    cout << "n1 /= 9.0 : " << n1.div(9.0) << endl;
    cout << "n2 = n1 : " << n2.setValue(n1.getValue());

    return 0;
}

```

- This code is almost the same as the previous exercise. But this time the class mostly used for its member functions. The member functions calculate arithmetic problems and assign the result to the only member variable from the class. This is unique code because the values are assigned to only one variable.

```

n1 : 0
n1 += 10.5 : 10.5
n1 -= 1.5 : 9
n1 *= 3.0 : 27
n1 /= 9.0 : 3
n2 = n1 : 3

```